Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/11/2025
PI Name:	Huang, Alex M.D., Ph.D.		
Project Title:	Exercise Countermeasure to Prevent Ocular Structural and Functional Changes in a Terrestrial Model of SANS (80NSSC22K1803)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	The PI moved from Doheny Eye Institute to University of California, San Diego in 2023.		
Project Type:	Ground	**Solicitation / Funding Source:**	2018-2019 HERO 80JSC018N0001-SANS: Spaceflight Associated Neuro-ocular Syndrome Countermeasures. Appendix C
Start Date:	08/22/2022	**End Date:**	08/21/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	7
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/21/2026 per V. Lehman/NASA JSC (Ed., 10/8/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Laurie, Steven (KBR./NASA Johnson Space Center) Lee, Stuart (KBR/NASA Johnson Space Center) Macias, Brandon (NASA Johnson Space Center) Marshall-Goebel, Karina (KBR/NASA Johnson Space Center) Sadda, Srinivas (Doheny Eye Institute) Linda, Loerch (NASA Johnson Space Center) Young, Millennia (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC22K1803		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Ed. Note (3/30/23): Project continues "Exercise Countermeasure to Prevent Ocular Structural and Functional Changes in a Terrestrial Model of SANS," Grant #80NSSC20K1034 with original Period of Performance 07/01/2020-06/30/2023. See that project for previous reporting. Optic disc edema develops in about 16% of astronauts during long-duration spaceflight and is of high concern to the NASA medical community and a target of therapeutic treatment given risk of vision loss. Currently, there is no reliable method to predict which crewmembers will develop disc edema. Moreover, it was previously believed that only the spaceflight environment could produce optic disc edema in normal healthy subjects. Recently our research team developed a novel ground-based spaceflight analog that reproduced disc edema in healthy test subjects that can be used to test novel pathophysiological hypotheses and possible Spaceflight Associated Neuro-ocular Syndrome (SANS) countermeasures. The research outlined in this proposal will use this new ground-based spaceflight analog advanced by NASA to elucidate the structural and functional impact of optic disc edema and to evaluate alterations to retinal and optic nerve blood flow to understand their contributions to the etiology of SANS. Further, a novel countermeasure will be tested to prevent the development of disc edema, functional decline in ganglion cell function, and vascular alterations associated with this ground-based spaceflight analog. This proposal will (1) determine combined structural and functional ocular alterations caused by the development of optic disc edema in this spaceflight analog, (2) determine the role of altered vascular blood flow in the development of optic disc edema in this spaceflight analog, and (3) determine if daily aerobic exercise in combination with veno-occlusive thigh cuffs can be used as a preventative countermeasure against the formation of optic disc edema. Thus, this proposal utilizes a newly developed spaceflight analog and will allow us to test novel hypotheses for predicting, characterizing, and preventing the development of optic disc edema.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Outcomes of this research can benefit humans in space and on Earth. First, SANS is a hurdle to long-haul space travel. The ocular alterations in SANS can impact vision, which represents a major safety concern during spaceflight. Thus, better understanding SANS and developing countermeasures to mitigate SANS improves the safety of space travel. Then, the major pathophysiological hypothesis for why SANS is occurring is the idea of fluid shifts. This concept states that gravity normally pulls total body fluid to the lower extremities of humans, and without gravity this fluid re-distributes to the head. It is hypothesized that the increased volume in the head leads to SANS. Increased volume in the head can also lead to other clinical manifestations such as a sensation of facial fullness and increased intraocular (eye) or intracranial (brain) pressure. The latter two are important in many Earth-bound diseases. For example, in glaucoma, head-ward fluid shifts are known to increase eye pressure which can lead to blindness. Thus, better understanding fluid shifts and how to minimize them may lead to additional benefits in the treatment of Earth-bound diseases.
Task Progress:	Specific Aims Specific Aim 1: To determine if donning veno-occlusive thigh cuffs during and for 120 minutes after daily aerobic exercise can be used to (a) reverse headward fluid shift from bedrest for a sustained period of time and (b) prevent ocular structural and functional changes from developing at the end of bedrest. Specific Aim 2: To determine the role of altered vasculature in the development of optic disc edema in this spaceflight analog. Supplemental Specific Aim: To develop artificial intelligence (AI) models to study SANS. Materials and Methods This study occurred at the :envihab facility in the German Aerospace Center (DLR). This study was a part of a larger head-down tilt bedrest (HDTBR) Countermeasure (CM) study. There were 4 campaigns and 4 conditions, each with ~12 subjects. The four arms of the study included (1) a 6-hour upright control and (2) a 30-day bedrest 6-degree head-down tilt (HDT) control. All experimental arms included the 30-day 6-degree bedrest HDT condition combined with two CMs: (1) veno-occlusive thigh cuffs and exercise (EX+CUFF) and (2) lower-body negative pressure (LBNP). The CM focus of this proposal is the EX+CUFF. For this proposal, the exercise and veno-occlusive thigh cuff CM were performed 6 days a week. Each session included exercise (60 min) that was sustained at ~60% of pre-bedrest peak oxygen consumption with the veno-occlusive thigh cuff applied at 50 mm Hg for six hours after exercise cessation. This length of time was increased over the original plan to better match inflight data. Endpoints were measured before, during, and after the 30-day bedrest period. These included optical coherence tomography (Spectralis HRA+OCT) of the eye, optical coherence tomography angiography (OCTA) of the eye, and electrophysiology (ERG) for ganglion cell function. Intravenous fluorescein angiography (IVFA) was performed to assess blood:brain vascular integrity at the optic nerve head immediately after the bedrest. Then, given the large amount of OCT data in HDTBR as well as after flight, AI models were trained. The first model was trained using ground truth identification of various landmarks in the eye to improve and automate the measurement of optic disc edema. The second model was trained using baseline imaging and the knowledge of whether an individual developed optic disc edema during flight or HDTBR. The last model was trained using baseline imaging and the knowledge of how much the nerve changed during flight or HDTBR. The last two models were tested using pre-flight or pre-BR images it had never seen before to predict whether the test subject would get optic disc edema and by how much. Results Optic nerve edema was seen in some cases. Only the upright CM, which fully reversed headward fluid shifts for a period of time, resulted in an average change to the optic nerve that was less than the NASA threshold for SANS. The remaining groups all showed continued optic nerve edema with no differences between groups. As a result, all subjects were then combined. Previously, astronaut research data had shown that a baseline small optic cup was a risk factor for optic disc edema. We evaluated the same endpoint in this study and showed the same result. There was a statistically significant relationship between a smaller baseline optic cup and increasing optic disc edema. This presence of risk factors was supported by the AI models. After training, the models could predict the development of SANS using baseline images it had not seen before with moderate success. The models relied on the retinal nerve fiber layer (RNFL), regions around the retinal pigment epithelium (RPE), the choroid near the BMO, and the deep optic nerve to make its determination.

The combined subjects were then evaluated based on optic nerve blood vessel parameters. We found that as subjects underwent HDTBR, the optic nerve vessel density increased on the optical coherence tomography angiography (OCTA). Then there was a statistically significant but mild decline in vessel density that was associated with increasing optic disc edema. Qualitatively, some eyes showed a normal optic appearance on intravenous fluorescein angiography (IVFA). However, other eyes demonstrated nerves that had vessel leakage. Quantitatively, a background-adjusted and vessel-subtracted average fluorescence intensity was measured for the optic nerve, showing a statistically significant increase in signal or leakage associated with increasing optic disc edema.

Discussion

In this study, we tested headward fluid shift CMs and studied the blood vessels around the optic nerve in a 30-day strict HDTBR model of SANS. The result confirmed that HDTBR leads to SANS-like optic nerve head (ONH) swelling, presenting with variable responses across individuals similar to what occurs in astronauts during spaceflight. Then, we showed for the first time that intermittent full reversal of headward fluid shifts from upright body positioning resulted in the smallest degree of ONH swelling seen to date. This result is important because it provides strong support for the fluid-shift hypothesis. The exercise and cuff and lower-body negative pressure CMs, however, did not appear to prevent ONH swelling. Therefore, all of the subjects were combined to evaluate the relationship of the optic disc cup and optic nerve blood vessels to ONH swelling. Given the variable response of the subjects to HDTBR, we considered that there may be SANS risk factors of disease propensity. Our result that a smaller baseline optic disc cup is associated with greater ONH swelling not only supports this but replicates the same result from astronauts. This was further supported by AI models that could predict the occurrence and magnitude of optic disc edema using baseline imaging only. Consistent with the optic cup data above, what this means is that baseline structural features (that an AI model can appreciate) put individuals at greater risk of optic disc edema, either during flight or BR and these factors need to be better understood. Then, our evaluation of the ONH vasculature showed that the changes to the ONH have functional consequences. After HDTBR, there is decreased blood flow perfusion density to the nerve, and there is leakage at those blood vessels. While the changes were mild, this study only involved a 30-day HDTBR period. Translated to astronauts and potentially longer missions (such as a mission to Mars, which could take years), the potential for damage to the blood vessels around the optic nerve leading to optic nerve tissue damage is a significant concern. In conclusion, the results of this study continue to emphasize the need to have strategies to treat SANS, prevent SANS, and/or identify those at greatest risk for SANS before committing to long-haul space travel.

Bibliography Type:	Description: (Last Updated: 07/02/2026)
Articles in Peer-reviewed Journals	Svoronos AA, O'Grady CS, Walker E, Afshari NA, Macias BR, Laurie SS, Weinreb RN, Huang AS. "Analysis of spaceflight-associated biometric and refractive changes in astronauts." Am J Ophthalmol. 2025 Aug;276:146-56. http://dx.doi.org/10.1016/j.ajo.2025.04.001 ; PMID: 40194644 , Aug-2025
Articles in Peer-reviewed Journals	Huang AS, Jalili J, Walker E, Weinreb RN, Laurie SS, Macias BR, Christopher M. "Artificial intelligence deep learning models to predict spaceflight associated neuro-ocular syndrome (SANS)." Am J Ophthalmol. 2025 Oct;278:115-123. https://doi.org/10.1016/j.ajo.2025.06.009 ; PubMed PMID: 40505849; PubMed Central PMCID: PMC12481839 , Oct-2025
Articles in Peer-reviewed Journals	He Y, Karanjia R, Zhang X, Wanderer D, Walker E, Lee SH, Poczatek MJ, O'Grady CS, Marion KM, Young M, Moestl S, Mulder E, Stern C, Sadda SR, Laurie SS, Macias BR, Huang AS. "Optic nerve vasculature and countermeasure assessment in a bedrest analogue of spaceflight-associated neuro-ocular syndrome." Am J Ophthalmol. 2025 Oct;278:317-327. https://doi.org/10.1016/j.ajo.2025.06.032 ; PubMed PMID: 40545016; PubMed Central PMCID: PMC12462775 , Oct-2025
Articles in Peer-reviewed Journals	Laurie SS, Lytle JR, Pickering SK, Poczatek MJ, Lee SMC, Pardon LP, Miller AE, Martin DS, Mulder E, Moestl S, Pesta D, Frett T, Stern C, Tank J, Hoenemann JN, Sadda S, Karanjia R, Marshall-Goebel K, Young M, Cole CW, Huang AS, Macias BR. "Daily use of countermeasures to prevent headward fluid shifts throughout 30 days of strict head-down tilt bedrest." J Appl Physiol (1985). 2025 Aug 1;139(2):394-413. https://doi.org/10.1152/japplphysiol.00635.2024 ; PMID: 40590841 , Aug-2025
Articles in Peer-reviewed Journals	Berger L, Mulder E, Zange J, Frings-Meuthen P, Frett T, Simon P, Hoenemann JN, Poczatek MJ, Laurie SS, Huang AS, Macias B, Jordan J, Tank J, Rittweger J, Schmitz MT, Pesta D. "Aerobic exercise and subsequent venoconstrictive thigh cuffs preserve cardiorespiratory fitness but not muscle function after 30 days of bedrest." Med Sci Sports Exerc. 2025 Dec. 57(12):2887-901. http://dx.doi.org/10.1249/MSS.0000000000003803 ; PubMed PMID: 40605173 , Dec-2025
Conference Materials (Downloadable)	Huang, A.S., He, Y., X. Zhang., Wanderer, D., Lee, S.H., Poczatek, M., Walker, E., Karanjia, R., Lee, S.M.C., Young, M., Marion, K.M., Mulder, D., Stern, C., Sadda, S.R., Laurie, S.S., and Macias, B.R. "Exercise and Lower Extremity Thigh Cuff Countermeasures to Ocular Changes During Strict Head-down Tilt Bedrest: The SANS Countermeasure Bedrest Study. " 2025 NASA Human Research Program Investigators Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 SANSA IWS2025.SANS.CM.pdf
Conference Materials (Downloadable)	Svoronos AS, O'Grady C, Walker E, Macias BR, Laurie SS, Afshari NA, Brunstetter TJ, Weinreb RN, Huang AS. "Characterization of spaceflight-associated changes in refraction and ocular biometry in astronauts." 2025 NASA Human Research Program Investigators Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 SANSRefractiveChangesHRPIWS.FINAL.pdf
Conference Materials (Downloadable)	Christopher M, Jalili J, O'Grady C, Weinreb RN, Laurie SS, Macias BR, Huang AS. "Artificial intelligence predicts quantitative retinal thickness changes and SANS onset during spaceflight. Artificial intelligence predicts quantitative retinal thickness changes and SANS onset during spaceflight." 2025 NASA Human Research Program Investigators Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 AI-IWS2025-abstract-FINAL.pdf

Conference Materials (Downloadable)	Wanderer D, He Y, Zhang X, Lee SH, Poczatek MJ, O'Grady C, Karanjia R, Lee SMC, Marion K, Mulder E, Stern S, Sadda SR, Laurie SS, Macias BR, Huang AS. "Optic nerve vasculature and countermeasure assessment in a ground-based analogue of spaceflight-associated neuro-ocular syndrome (SANS)." Association for Research in Vision and Ophthalmology (ARVO) Annual Meeting, Salt Lake City, Utah, May 4-8, 2025. , May-2025 Wanderer.ARVO.FINAL.pdf
Conference Materials (Downloadable)	Huang AS, Jalili J, O'Grady C, Park J, Weinreb RN, Laurie SS, Macias BR, Christopher M. "Artificial intelligence predicts development of spaceflight-associated neuro-ocular syndrome (SANS) from pre-flight data." Association for Research in Vision and Ophthalmology (ARVO) Annual Meeting, Salt Lake City, Utah, May 4-8, 2025. , May-2025 Huang.ARVO.2025.FINAL.pdf